Touhidul Alam Seyam

Software Engineer | Web, App, Backend, ML/AI

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PROFESSIONAL SUMMARY

Software engineer, full-stack developer, and applied AI researcher who turns technical ideas into production-ready products and reproducible experiments. Professional experience since 2021 across TypeScript, Python, React, Next.js, Node.js, backend services, data systems, and machine learning. Builds client applications, open-source developer tools, AI agents, and research software; author or co-author of ten source-backed research works.

CORE SKILLS

Languages: TypeScript, JavaScript, Python, SQL, Go, Zig

Product Engineering: React, Next.js, Node.js, Django, FastAPI, REST APIs, React Native, Tailwind CSS

Data and Infrastructure: PostgreSQL, MongoDB, Redis, SQLite, Prisma, Convex, Docker, Git, AWS, Vercel

AI and Research: scikit-learn, TensorFlow, PyTorch, XGBoost, LightGBM, computer vision, agent evaluation, reproducible experiments

PROFESSIONAL EXPERIENCE

Software Engineer | Agentic Institute | **Remote | Dec 2025-Present**

- Contribute to application development and AI-enabled, agentic software systems.
- Work across implementation, integration, testing, and delivery using modern web and backend technologies.

Software Engineer | Hello World Communications Ltd | **Chattogram, Bangladesh | Aug 2024-Present**

- Build, review, deploy, and maintain full-stack web applications for client and internal use across frontend, backend, database, and AI-integration layers.
- Collaborate with cross-functional stakeholders on requirements, implementation, production releases, and ongoing application support.

Freelance Developer | Self-employed | **Remote | Mar 2021-2025**

- Delivered web applications from requirements and interface design through backend implementation, deployment, and ongoing support.
- Built digital products for small businesses, nonprofit and academic organizations, and event teams using modern JavaScript and Python stacks.

SELECTED PROJECTS

bun-scikit | **TypeScript, Bun, Zig** | **github.com/Seyamalam/bun-scikit**

- Built a scikit-learn-inspired machine-learning library with native Zig acceleration, tests and benchmark gates, 209 tracked runtime exports, and documented APIs across preprocessing, model selection, ensembles, clustering, and metrics.

RoboFusion | **Python, FastAPI, WebSockets, ESP32** | **github.com/Seyamalam/Robofusion**

- Developed a multi-hazard campus response platform with device integration, authentication, persistence, incident-lifecycle management, real-time dashboards, automated tests, and load-testing tools.

ASRRO Portal | **Next.js, TypeScript, Convex, Better Auth** | **github.com/Seyamalam/asrro**

- Developed a public website and role-aware organization-management system covering membership approvals, events, attendance, content, reports, notifications, and restricted finance workflows.

Kaggriculture Autonomous Agent | **Python, Simulation, Evaluation** | **github.com/Seyamalam/Kaggriculture**

- Engineered a deterministic farming agent and evaluation harness with seeded seat-swapped tournaments, frozen replay corpora, manifest hashing, and regression gates.

EDUCATION

B.Sc. (Hons.) in Computer Science and Engineering | BGC Trust University Bangladesh **| Chattogram, Bangladesh | Expected Dec 2026**

Currently in the 8th semester. Focus: software engineering, machine learning, computer vision, applied AI, and research-driven product development.

SELECTED RESEARCH

- *“Architectures for AI-Driven Visual Assistance: Evaluating Server-Mediated Mobile and Direct Access Desktop Client Implementations.”* Lecture Notes in Networks and Systems, 2026. DOI: 10.1007/978-3-032-15764-5_45
- *“AgriScan: Next.js powered cross-platform solution for automated plant disease diagnosis and crop health management.”* Journal of Electrical Systems and Information Technology, 2024. DOI: 10.1186/s43067-024-00169-7
- *“Next-Generation K-Means Clustering: Mojo-Driven Performance for Big Data.”* International Journal of Intelligent Information Systems, 2025. DOI: 10.11648/j.ijiis.20251401.12

SELECTED CERTIFICATIONS

IBM Applied Data Science with Python; IBM Deep Learning with TensorFlow; Redis for Python Developers; Cisco Python Essentials 1 and 2; HackerRank Problem Solving (Intermediate)